

Navigator 1

Autonomous Robotics Platform

Engineered by **Team Robonavigators** — a fault-tolerant, AI Like-assisted rover powered by a custom embedded OS, untethered web interface, and six intelligent driving modes.

Built From the Ground Up

Navigator 1 bypasses off-the-shelf kits entirely. At its core is a **custom 884-line C++ embedded operating system** running on the Arduino UNO R4 WiFi — delivering real-time control, autonomous decision-making, and a standalone asynchronous web server in a single, crash-proof machine.

884 Lines of C++

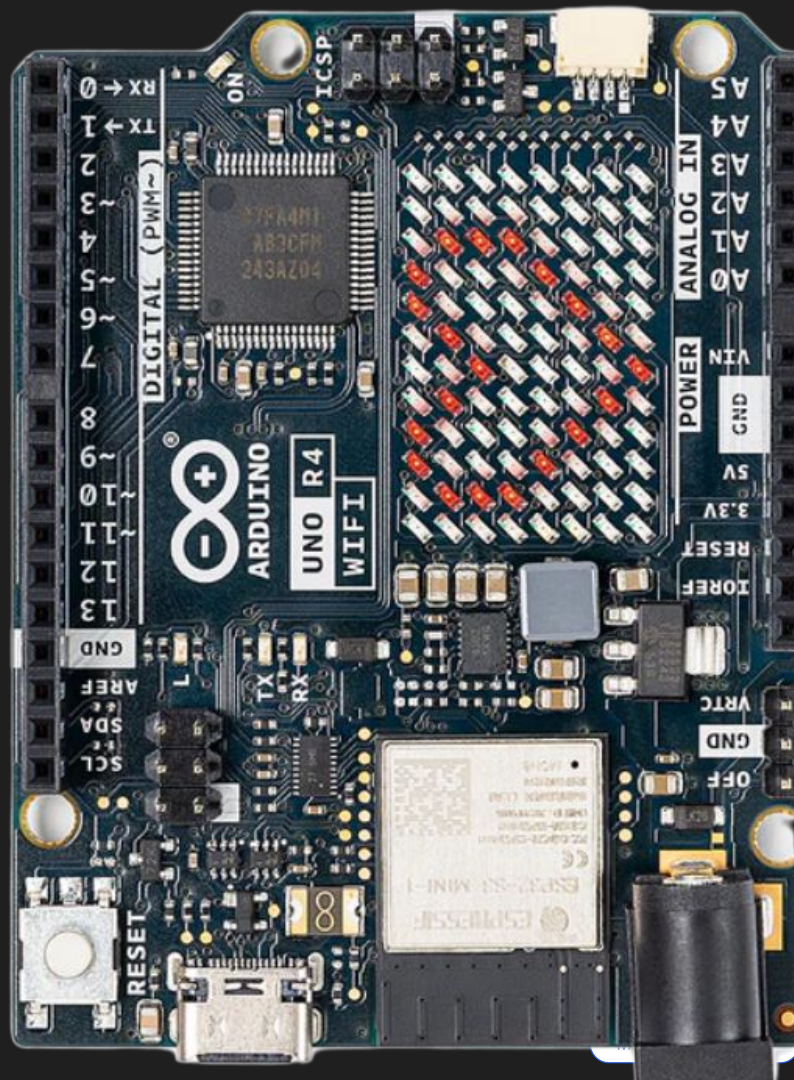
Custom embedded OS, zero dependencies

Arduino UNO R4 WiFi

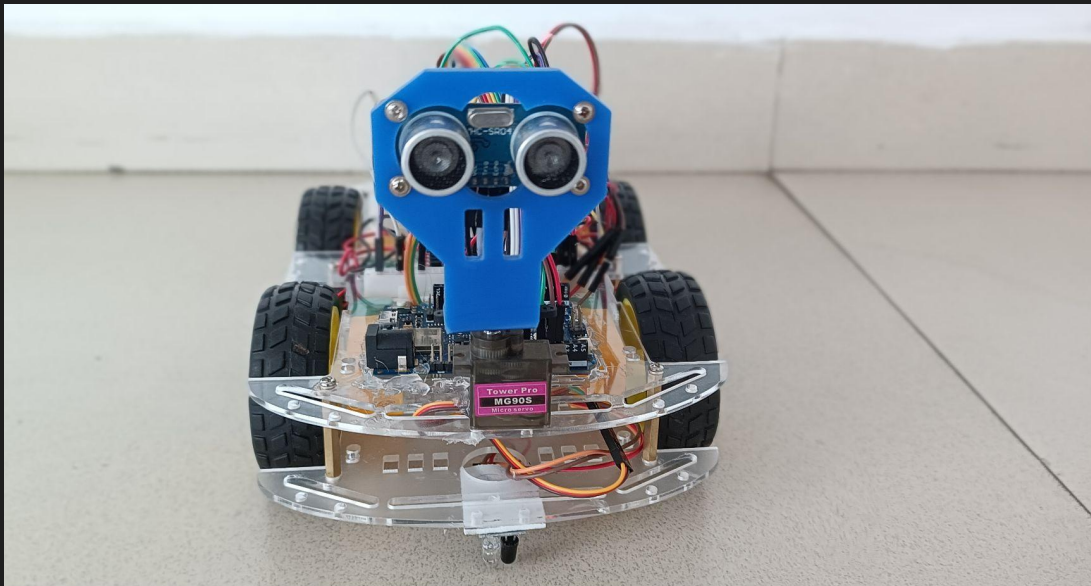
Native Wi-Fi, onboard LED matrix

Standalone Web Server

Asynchronous, zero-reload interface



Hardware & Sensor Architecture



Advanced Sensor Array

- **Front & rear ultrasonic sensors** for full 360° spatial mapping
- **Downward-facing IR sensor** for cliff detection and drop-off prevention
- **180° servo-mounted radar** for sweeping obstacle scans in Autopilot mode

Custom Charging Matrix

A proprietary **4-bay TP4056 independent charging station** ensures perfectly balanced 18650 lithium-ion cells — eliminating the risks of series charging entirely.

Untethered Web OS

No laptop. No monitor. No tether. Navigator 1 boots into a fully functional web interface accessible from any smartphone or tablet.

Zero-Lag Dashboard

Custom HTML/JS interface uses **asynchronous fetch requests** — commands execute and telemetry updates without ever reloading the page.

PC-Free Boot Sequence

A physical hardware switch toggles AP/STA networks. The onboard **LED matrix scrolls the assigned IP address** — no external monitor needed.

Network Failsafes

A **4-second heartbeat** catches dropped connections instantly. An **8-second boot timeout** safely defaults to offline autonomous mode if Wi-Fi is unavailable.

Physics & Active Safety Engine



Smart Active Braking

Injects a **50ms reverse-torque pulse** to lock the drivetrain and kill skidding momentum — replacing standard coasting.



Dual-Sensor Active Shield

A permanent **25cm virtual bumper** front and rear. If breached, the OS overrides human input to prevent hardware damage.



Interrupt-Level Cliff Detection

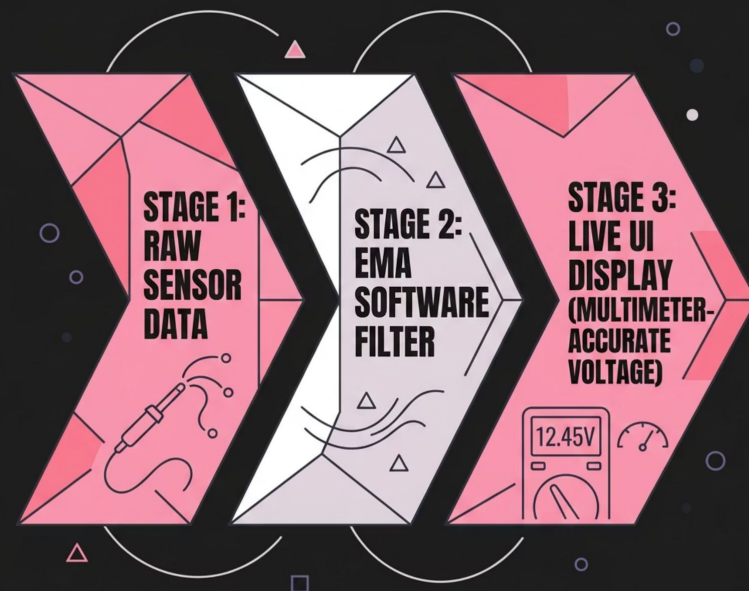
Downward IR sensors trigger **immediate active braking** at table edges — rendering the rover effectively un-fallable.

Live Telemetry & Precision Monitoring

EMA-Filtered Battery Readings

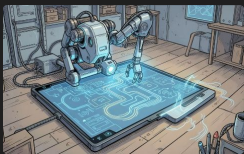
Navigator 1 employs an **Exponential Moving Average (EMA) software filter** to deliver live, multimeter-accurate battery voltage readings directly to the web UI — smoothing out noise without sacrificing responsiveness.

-  The EMA filter continuously weights recent samples more heavily, providing a stable voltage readout even under dynamic motor load changes.



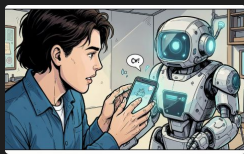
Six Intelligent Operating Modes

Navigator 1 ships with six distinct driving modes — each protected by the Active Safety Engine in the background.



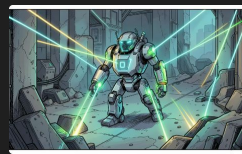
Draw & Follow

Translates touchscreen drawings into physical floor vectors with auto-pause on obstacles.



Voice Control

Browser Web Speech API integration for zero-latency continuous voice commands.



Autopilot

Radar-assisted 180° scanning calculates and executes optimal avoidance paths.

Advanced Modes: Tracking & Remote



Object Follow

Dynamic spatial tracking locks onto targets while maintaining a safe **20cm–80cm** standoff distance.



Smart Idle Shield (Default)

An active parking guard that safely retreats from frontal proximity — checking rear sensors first to avoid backing into walls.



Remote Control

Precision digital joystick navigation with the **Active Shield always engaged** in the background for collision-free manual driving.

Platform Capabilities at a Glance

884

Lines of C++

Custom embedded OS

6

Smart Modes

Each protected by the Active Safety Engine

25cm

Virtual Bumper

Dual-sensor Active Shield range

50ms

Brake Pulse

Reverse-torque injection for instant stop

Navigator 1 – Ready to Deploy

A fully autonomous, crash-proof, untethered robotics platform — engineered without shortcuts, powered by custom software, and ready for real-world operation.

Custom OS

884-line C++ embedded system

Active Safety

Physics-based braking & shielding

Web Interface

Zero-lag, PC-free operation

6 Modes

Draw, Voice, Autopilot, Follow, Idle, RC

